Problem Set VII: Edgeworth Box, Robinson Crusoe

Paolo Crosetto

Exercises will be solved in class on *March 22nd, 2010*

1. Edgeworth box

Consider a pure-exchange, private-ownership economy, consisting in two consumers, denoted by $i = 1, 2$, who trade two commodities, denoted by $l = 1, 2$. Each consumer i is characterized by an endowment vector, $\omega_i \in \mathbb{R}^2_+$, a consumption set, $X_i = \mathbb{R}^2_+$, and regular and continuous preferences, $\succsim_i$ on X_i. Initial endowments are given by $\omega_1 = (4, 2)$ and $\omega_2 = (2, 3)$. Individual utility functions are $u_1(x_{11}, x_{21}) = x_{11}x_{21}$ and $u_2(x_{12}, x_{22}) = x_{12} + x_{22}$.

1. Draw the Edgeworth Box for this economy, drawing endowment point ω and the indifference curves passing through it for both consumers.
2. Find analytically the Pareto Set (interior points and, separately, boundary points) and the Contract curve. Draw them both in the Edgeworth Box.
3. Find the competitive equilibrium prices and alocations. Draw it in the Edgeworth Box.
4. Take the point $\tilde{x} = ((\tilde{x}_{11}, \tilde{x}_{21}), (\tilde{x}_{12}, \tilde{x}_{22})) = ((1, 1), (5, 4))$. Draw it in the Edgeworth box, show that it is in the Pareto set, and write down the implicit shadow prices. Compute the transfers T_1 and T_2, with $T_1 + T_2 = 0$, that ought to be assigned to either consumer in order that, starting from the initial endowments, the allocation concerned can be obtained as a price equilibrium with transfers.

Setting up the box

- The dimension of the box are given by $\bar{\omega}_1 = 4 + 2 = 6$ and $\bar{\omega}_2 = 2 + 3 = 5$;
- The indifference curves are level sets of the utility functions:

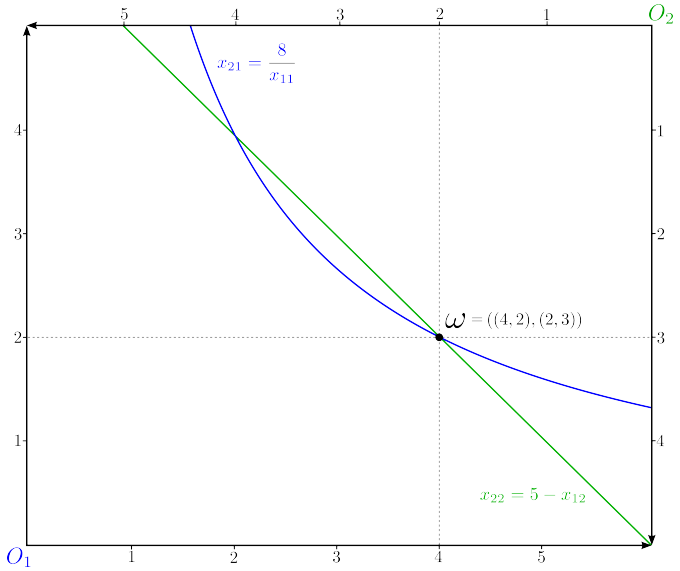
$$\text{For 1: } x_{11}x_{21} = u_1(\omega) \qquad \text{For 2: } x_{12} + x_{22} = u_2(\omega)$$

- The utility level of endowment is $u_1(\omega) = 4 \cdot 2 = 8$ for consumer one and $u_2(\omega) = 2 + 3 = 5$ for consumer two;
- Hence, the indifference curves passing through the endowment point ω are:

$$\text{For 1: } x_{21} = \frac{8}{x_{11}} \qquad \text{For 2: } x_{22} = 5 - x_{12}$$

- Note that consumer 1 has Cobb-Douglas preferences, while consumer 2 has perfect substitutes;
- See the drawing below for graphical representation.

Edgeworth Box and indifference curves, graphics



Pareto Set, interior points

- The interior points of the Pareto Set are given by all the points in which the two MRS^i_{12} are equal.

$$MRS^1_{12} = \frac{\partial u_1}{\partial x_{11}} \left(\frac{\partial u_1}{\partial x_{21}} \right)^{-1} = \frac{x_{21}}{x_{11}}$$

$$MRS^2_{12} = \frac{\partial u_2}{\partial x_{12}} \left(\frac{\partial u_2}{\partial x_{22}} \right)^{-1} = \frac{1}{1} = 1$$

- Hence, the interior points for player one are given by the condition

$$x_{21} = x_{11} \quad \text{w.r.t.} \, O_1$$

- and, for player two, by

$$5 - x_{22} = 6 - x_{12} \quad \Rightarrow \quad x_{22} = x_{12} - 1 \quad \text{w.r.t.} \, O_2$$

Pareto Set, boundary points, Contract Curve

- There are boundary points belonging to the Pareto Set: the Pareto Set always starts from the two origins of the EB.
- Hence all points for which $x_{22} = 0$ and $x_{12} \leq 1$ are in the Pareto Set.
- This is because at those points, consumer 2 would like to sell more of good 2, but he can't, while consumer 1 is willing to buy more of both goods, but he can buy more only of good 1 since he holds already all of good 2.
- See the graphics for intuition.
- The contract curve is all the points in the Pareto Set for which utility is higher or equal than the utility given by the endowments.
- Hence, for player one we solve

$$\begin{cases} x_{11} x_{21} = 8 \\ x_{21} = x_{11} \end{cases} \quad \Rightarrow \quad x^2_{11} = x^2_{21} = 8 \quad \Rightarrow \quad x_{11} = x_{21} = 2\sqrt{2}$$

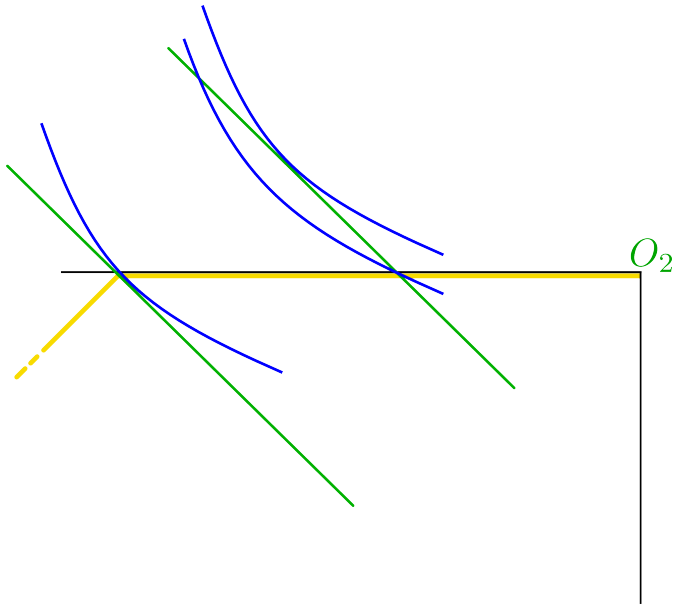
- And for player 2 we solve

$$\begin{cases} x_{12} + x_{22} = 5 \\ x_{22} = x_{12} - 1 \end{cases} \quad \Rightarrow \quad 2x_{12} = 6 \quad \Rightarrow \quad x_{12} = 3; \quad x_{22} = 3 - 1 = 2$$

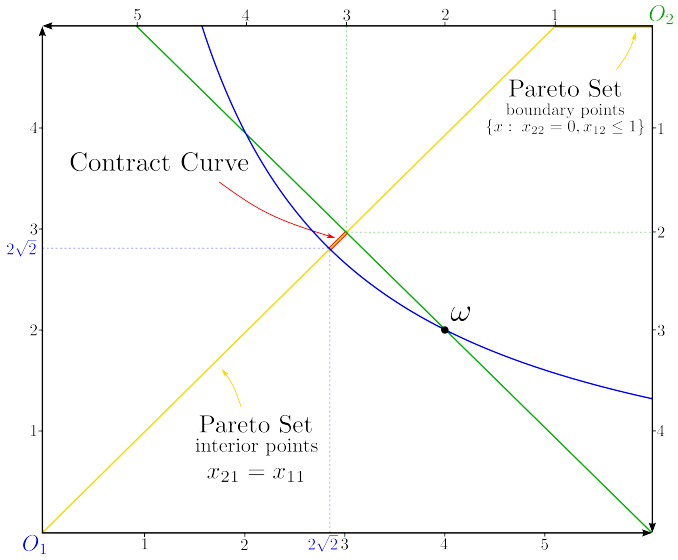
- The Contract curve, making reference to the origin of consumer one, is:

$$\{x \in \text{Pareto Set} : \; 2\sqrt{2} \leq x_{11} \leq 3 \, \text{and} \, 2\sqrt{2} \leq x_{21} \leq 3\}$$

Pareto Set, boundary points, graphics



Pareto Set, Contract Curve, graphics



Equilibrium

- In equilibrium, both players optimize given prices, and prices move to guarantee market clearing.
- Exploiting MRS calculations, for consumer one, imposing $p_1 \equiv 1$:

$$MRS_{12}^1 = \frac{p_1}{p_2} \quad \Rightarrow \quad \frac{x_{21}}{x_{11}} = \frac{1}{p_2}$$

$$MRS_{12}^2 = \frac{p_1}{p_2} \quad \Rightarrow \quad 1 = \frac{1}{p_2} \quad \Rightarrow \quad p_2^* = 1$$

- Plugging the equilibrium price system $p^* = (1, 1)$ into the budget constraints we get demand functions for player 1 and 2.
- For consumer 1:

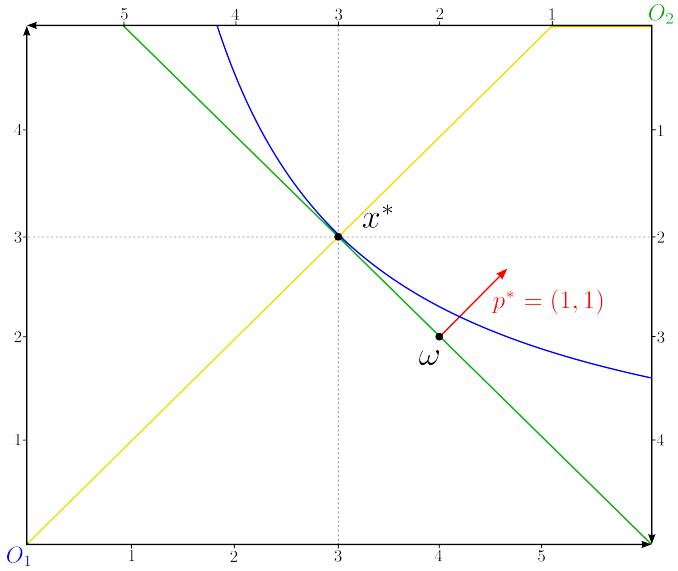
$$\begin{cases} x_{11} = x_{21} \\ x_{11} + x_{21} = 6 \end{cases} \quad \Rightarrow \quad x_{11}^* = x_{21}^* = 3$$

- and, for consumer 2:

$$\begin{cases} x_{22} = x_{12} - 1 \\ x_{12} + x_{22} = 5 \end{cases} \quad \Rightarrow \quad x_{12}^* = 3;\ x_{22}^* = 2$$

- Hence, equilibrium is $x^* = ((3,3),(3,2))$, and $p^* = (1,1)$.

Edgeworth Box equilibrium, graphics



Price equilibrium with transfers

- An allocation in the Edgeworth Box can be sustained as a *Price Equilibrium with Transfer*;
- That is, a planner can make transfers and then let the consumers freely exchange, in order to lead them to a desired solution
- In this case the planner wants to support as an equilibrium the allocation $\tilde{x} = ((1,1),(5,4))$
- The planner must hence use transfers T_1, T_2 s.t.: $T_1 + T_2 = 0$

1. The point is in the Pareto Set: $\tilde{x}_1$ satisfies $x_{11} = x_{21}$ (and hence it satisfies the condition for the other consumer, too, no need to check).
2. The implicit shadow prices are constant in all of the Pareto Set, and were calculated to be $\tilde{p} = p^* = (1,1)$.
3. The Transfer must be calculated by comparing the value of endowment and the value of allocation $\tilde{x}$ for both consumers at prices $p^* = (1,1)$.

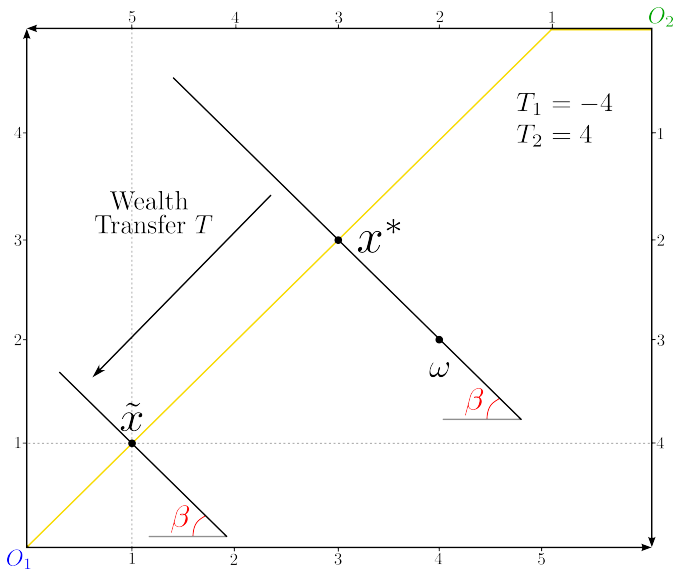
Calculating transfers

- A planner must transfer wealth in order to match for the difference in value at prices $p^* = (1,1)$ between the endowments and the preferred point, in this case $\tilde{x} = ((1,1),(5,4))$

$$\text{For 1: } p^* \cdot \tilde{x} = (1,1)\cdot(1,1) = 1+1 = 2 \quad p^* \cdot \omega_1 = (1,1)\cdot(4,2) = 4+2 = 6$$
$$\text{For 2: } p^* \cdot \tilde{x}_2 = (1,1)\cdot(5,4) = 5+4 = 9 \quad p^* \cdot \omega_2 = (1,1)\cdot(2,3) = 2+3 = 5$$
$$T_1 = p^* \cdot \tilde{x}_1 - p^* \cdot \omega_1 = 2-6 = -4; \quad T_2 = p^* \cdot \tilde{x}_2 - p^* \cdot \omega_2 = 9-5 = 4;$$

$$T_1 = -4; \quad T_2 = 4; \quad T_1 + T_2 = 0.$$

Price equilibrium with transfers, graphics



Recap: Robinson Crusoe Economy

The simplest possible general equilibrium model including production is one with two economic agents, 1 consumer and 1 producer, superimposed into one person only: lonely Robinson on his remote island.

- Note that price-taking behaviour is assumed for both producer and consumer;
- note as well that the consumer owns the firm and earns all of its profits.
- These two assumptions do not make much sense in 1-person economies, but the model is just an extreme simplification.

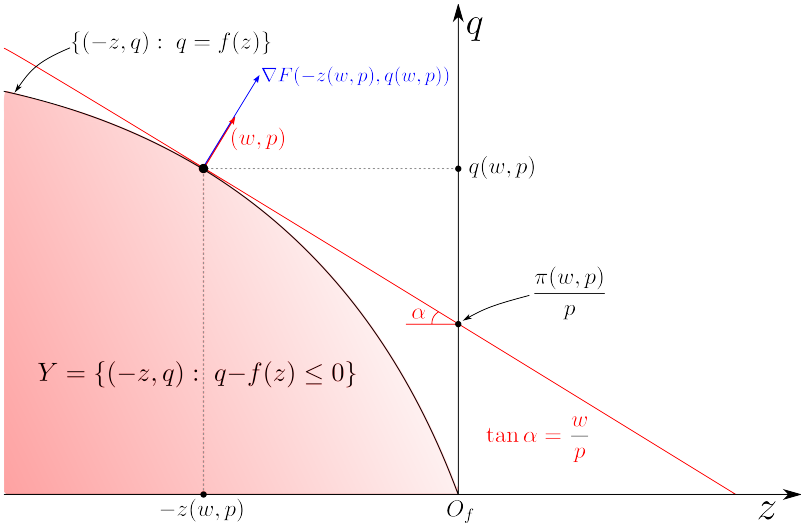
Recap: Robinson's producer problem

- Robinson as a producer acts as a competitive firm: maximises profits given technology.
- He hence solves

$$\max_{z \geq 0} \; pf(z) - wz$$

- And from this maximisation problem derives the firm's labor demand $z(w,p)$, output $f(z(w,p)) = q(w,p)$, and profits $\pi(w,p)$.

Robinson as a producer



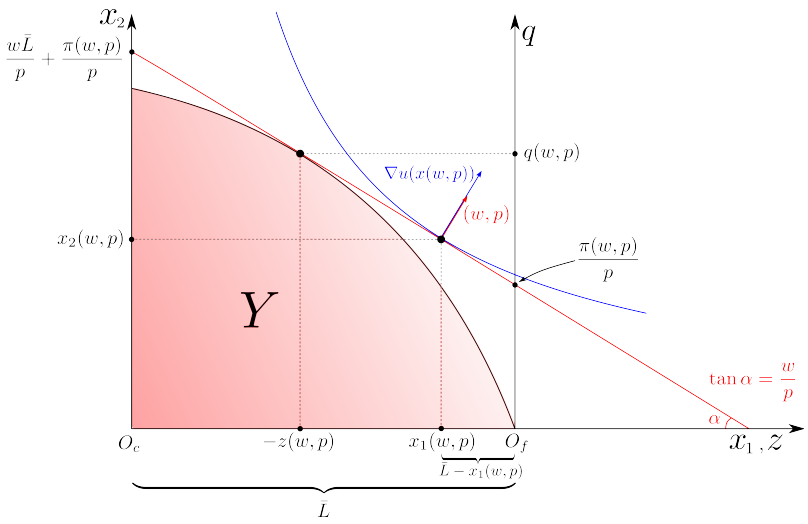
Recap: Robinson's consumer problem

- The consumer owns the firm: he hence solves a utility maximisation on two goods, leisure and a consumer good, having as resources his time ($\bar{L}$) and the profits generated by the firm.
- He hence solves

$$\max \quad u(x_1, x_2) \quad \text{s.t. } wx_1 + px_2 = \bar{L} + \pi(w, p)$$

- From which the two walrasian demands $x_1(w, p)$ and $x_2(w, p)$ can be derived,
- jointly with the labour supply function, $s(w, p) = \bar{L} - x_1(w, p)$.

Robinson as a consumer



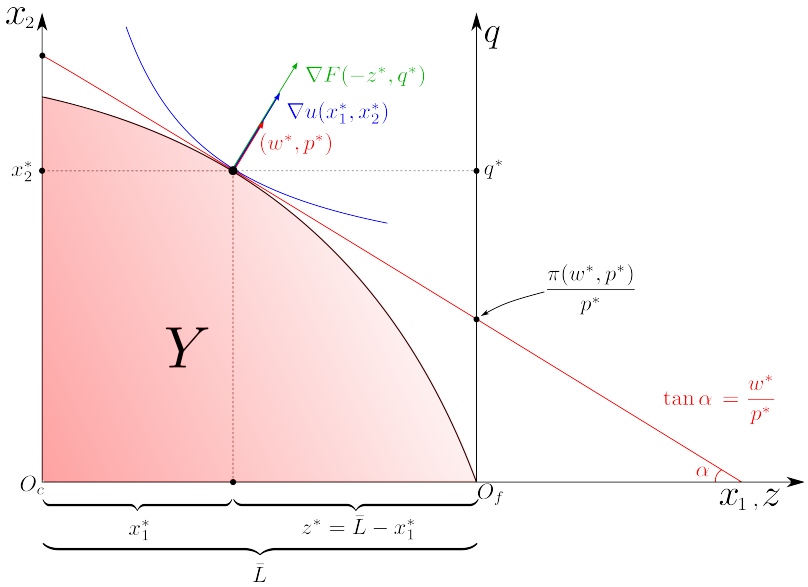
Recap: Robinsonian equilibrium

- An equilibrium is a price vector (w^*, p^*) at which both consumption and labour market clear,
- that is, at which

$$x_2(w^*, p^*) = q(w^*, p^*) \quad \text{and} \quad z(w^*, p^*) = \bar{L} - x_1(w^*, p^*)$$

- An equilibrium maximises utility subject to the technology and endowment constraints
- In this case, if functions are well-behaved, both theorems of welfare economics hold.

Robinsonian equilibrium



2. Robinson Crusoe Economy

Consider a "Robinson Crusoe economy", i.e., a private-ownership, competitive economy with only one consumer ($I = 1$), one producer ($J = 1$), and two commodities ($L = 2$). The producer is characterized by a single-output technology, with production set $Y = \{(-z, q) \in \mathbb{R}^2 \mid q - f(z) \leq 0 \text{ and } z \geq 0\}$, where z is the quantity of input ("labor time"), q is the quantity of output ("consumers' good"), and $f(z) = 2z$ is the production function. The consumer is characterized by a consumption set $X = \{x = (x_1, x_2) \in \mathbb{R}^2_+\}$, where x_1 and x_2 are the quantities of "leisure time" and "consumers' good", respectively, and a Cobb-Douglas utility function $u(x_1, x_2) = x_1^{\frac{2}{3}} x_2^{\frac{1}{3}}$. The consumer owns the endowments $\bar{\omega} = (\bar{L}, 0) \in \mathbb{R}^2_+$, where $\bar{L} = 24$ denotes the time units the consumer has at his disposal (time can be used as either "leisure time" or "working time"); the endowment of "consumers' good" is nil. The consumer owns the producer, thereby getting the whole of the latter's profits. Let p be the price of the "consumers' good" and w be the wage, i.e., the price of both the "leisure time" and the "working time". Let the "consumers' good" be the numeraire of the economy, and consequently set $p \equiv 1$.

1. After examining the properties of the production function, focussing particularly on the nature of the returns to scale, write down and solve the producer's profit maximization problem, determining the demand correspondence for "labor time" and the supply correspondence of the "consumers' good" (the only argument of both correspondences being the wage w, which coincides with the real wage, $\frac{w}{p}$, since $p \equiv 1$).
2. Find the producer's profit function $\pi(\cdot)$, showing in particular that, given the nature of the returns to scale, such function is not always well-defined: it can either take the value 0 or diverge to $+\infty$. Hence, prove that the value 0 is the only one to be consistent with a competitive equilibrium.
3. After examining the properties of the utility function, write down and solve the consumer-worker's utility maximization problem (for an interior optimum and assuming the producer's profits to be equal to 0, since this is the only value that has been proven to be consistent with a competitive equilibrium), thereby determining the Walrasian demand functions for both "leisure time" and "consumers' good" as well as the Walrasian supply function of "working time" (all having the wage w as the only argument).
4. Illustrate the producer's and the consumer-worker's problems in one and the same diagram.
5. Find the (unique) competitive equilibrium of the "Robinson Crusoe's economy" and plot it in the above diagram.
6. Prove that the competitive equilibrium allocation is the only Pareto-efficient allocation of the economy.

Properties of production function

- $f(z)$ shows *constant returns to scale*:
$$f(\alpha z) = 2\alpha z = \alpha f(z)$$
- The marginal product of labour is constant:
$$\frac{\partial f(z)}{\partial z} = 2$$
- Average product of labour is constant:
$$\frac{f(z)}{z} = 2$$
- We know that in the case of *CRS* the profit maximisation is not well defined.
- We will have to determine the equilibrium on the consumer side.

Profit maximisation problem,I

Robinson as a producer solves a profit maximisation problem:

$$\max_{z \geq 0} \quad f(z) - wz$$

And the Lagrangean can be set up to be:

$$\max \quad L(z, \lambda) f(z) - wz + \lambda(-z)$$

And the FOC for this problem (Kuhn-Tucker), are:

$$\begin{cases} 2 - w = \lambda \\ \lambda \geq 0 \\ \lambda z = 0 \end{cases} \Rightarrow \begin{cases} 2 \leq w \\ (2-w)z = 0 \end{cases}$$

That can be solved to yield

$$z(w) = \begin{cases} 0 & \text{for } w > 2 \\ \in [0, \infty) & \text{for } w = 2 \\ \infty & \text{for } w < 2 \end{cases}$$

Profit Maximisation Problem, II

The supply function $f(z(w))$ can then be calculated to be

$$q(w) = f(z(w)) = \begin{cases} 0 & \text{for } w > 2 \\ \in [0, \infty) & \text{for } w = 2 \\ \infty & \text{for } w < 2 \end{cases}$$

And the corresponding profit function can be calculated as usual by plugging the input demand correpsondence into the maximand:

$$\pi(w) = \max_{z \geq 0} \{pf(z) - wz\} = \max_{z \geq 0} \{2z - wz\} = \begin{cases} 0 & \text{for } w \geq 2 \\ \infty & \text{for } w < 2 \end{cases}$$

Infinite profits imply an infinite demand for labour. This is not compatible with general equilibrium, since the supply of labour by robinson-consumer cannot be infinte. Hence, the only value of profits that is consistent with a competitive equilibrium is $\pi(w) = 0$

Properties of utility function

This is a Cobb-Douglas utility function. It has the standard C-D properties:

$$u_1'(x_1, x_2) = \frac{\partial u(x_1, x_2)}{\partial x_1} = \frac{2}{3}\left(\frac{x_2}{x_1}\right)^{\frac{1}{3}}$$

$$u_2'(x_1, x_2) = \frac{\partial u(x_1, x_2)}{\partial x_2} = \frac{1}{3}\left(\frac{x_2}{x_1}\right)^{\frac{2}{3}}$$

$$MRS_12(x_1, x_2) = \frac{u_1'(x_1, x_2)}{u_2'(x_1, x_2)} = 2\frac{x_2}{x_1}$$

Hence, tha gradient is always positive, $\nabla u(x_1, x_2) \gg 0$, for $x \in \mathbb{R}^2_+$. Moreover, checking the bordered Hessian (do it!) we can conclude that the function is strictly quasi-concave.

Utility Maximisation problem, I

Robinson as a consumer solves a standard UMP, introducing in his budget constraint the maximal profits. In this case, $\pi(w) = 0$, so the problem, setting $p \equiv 1$, boils down to

$$\max_{x \geq 0} \; x_1^{\frac{2}{3}} x_2^{\frac{1}{3}} \quad \text{s.t. } wx_1 + px_2 = w\bar{L} + \pi(w) = 24w + 0 = 24w$$

FOCs boil down to

$$\begin{cases} MRS_{12}(x_1, x_2) = \dfrac{w}{p} \\ wx_1 + px_2 = w\bar{L} + \pi(w) \end{cases} \Rightarrow \begin{cases} 2\dfrac{x_2}{x_1} = w \\ x_2 = (24 - x_1)w \end{cases}$$

That can be easily solved to yield the demand for leisure $x_1(w)$ and the demand for consumer good $x_2(w)$:

$$x_1(w) = 16 \qquad x_2(w) = (24 - 16)w = 8w$$

The corresponding labour supply function, $s(w)$, can then be easily calculated:

$$s(w) = \bar{L} - x_1(w) = 24 - 16 = 8$$

Equilibrium

Equilibrium in the labour market implies:

$$s(w^*) = z(w^*) \quad \Rightarrow \quad z(w^*) = 8$$

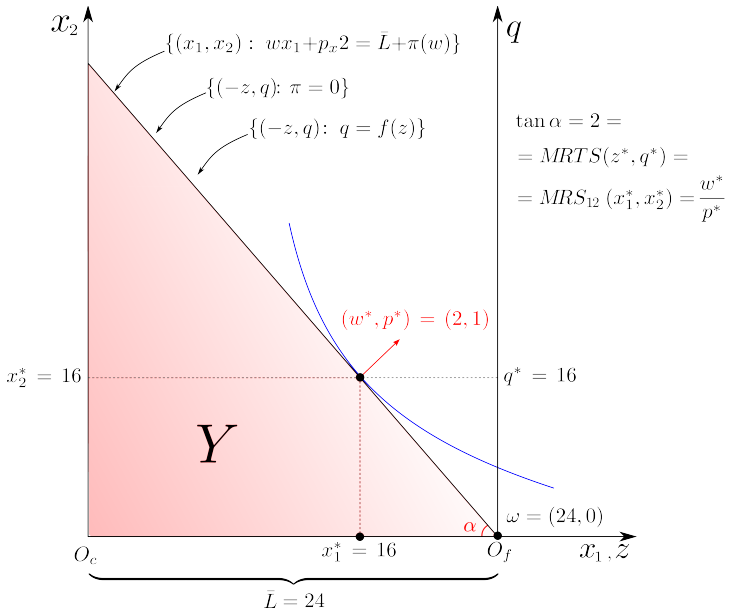
And equilibrium on the consumer good market in turn implies:

$$x_2(w^*) = q(w^*) \quad \Rightarrow \quad x_2(w^*) = 2(z(w^*)) = 16 = 8w^* \quad \Rightarrow \quad w^* = 2$$

Summing up, the equilibrium allocation and prices are given by:

$$E = (((x_1^*, x_2^*), (-z^*, q^*)), (w^*, p^*)) = (((16, 16), (-8, 16)), (2, 1))$$

Robinson Crusoe: graphics



The Walrasian equilibrium is the only Pareto Efficient allocation

Proof.

- In this case both the First and the Second Fundamental theorems hold
- Since the production set Y is a convex set,
- and the utility function is strictly quasi-concave.
- Hence, the set of the walrasian equilibria and the pareto set coincide
- And, the walrasian equilibrium being unique at E,
- it must also be the only Pareto Efficient allocation.

□